

Benign Safety Supervision Transfers, but the Jailbreak-Wrapper Advantage Is Presentation-Dependent

Anonymous ACL submission

Abstract

Hard-example safety tuning can make a model robust by making it reluctant, while jailbreak-shaped benign data reduces the resulting over-refusal. It remains unclear whether that benign data teaches transferable harmless behavior, wrapper-specific adaptation, or merely the effect of adding extra user context. An earlier analysis of our experiments concluded that benign-wrapper supervision produced presentation dependence without transfer. That conclusion rested on a wrapped-minus-dewrapped ($W - D$) contrast in which both arms already received benign supervision, so it could not establish the absence of transfer. We add the missing anchor: a normalization-matched harmful-only arm H_{norm} with identical hard harmful examples, optimizer slots, updates, initialization, and supervised-token denominator, but no benign text, labels, token IDs, or gradients. In this controlled same-core, same-target trial, benign supervision clearly transfers. On direct/XSTest presentations, $D - H_{\text{norm}}$ is +2.40 log-odds (95% CI [2.12, 3.25]), while $W - H_{\text{norm}}$ is +2.22 [1.99, 2.96]; raw direct safe/harmful fulfillment is 0.81/0.07 for D , 0.75/0.05 for W , and 0.215/0.000 for H_{norm} . A slot-placebo comparison rules out the matched loss denominator as the explanation. We therefore retract the stronger no-transfer interpretation. The defensible result is narrower: the incremental advantage of W over D is nonpositive on direct presentations and positive on wrapped presentations, even though both benign arms transfer substantially. This pattern does not identify a latent intent mechanism. Because cross-judge fulfillment agreement falls from 0.971 ($\kappa = 0.939$) on direct prompts to 0.788 ($\kappa = 0.453$) on wrapped prompts, we independently rescore a frozen matched audit with a qualified cross-family Llama judge. Transfer, the normalization control, and the presentation-dependent $W - D$ pattern reproduce under both automatic judges. Safety remains a strictly bounded negative:

fulfillment is not response harmfulness, no setting passes joint noninferiority, and we make no safety-improvement claim.

1 Introduction

Open-weight instruction-tuned language models are routinely adapted with lightweight supervised fine-tuning, and safety behavior is among the first properties such adaptation can alter (Ouyang et al., 2022; Bai et al., 2022; Qi et al., 2023). A practical recipe is *hard-example safety tuning*: mine prompts on which the target model’s own rollouts are frequently judged harmful, then fine-tune on its non-jailbroken responses to those prompts (Gupta et al., 2026). This substantially reduces jailbreak attack success. It also has a documented cost: the model becomes reluctant, refusing benign requests that merely resemble attacks in wording or presentation (Röttger et al., 2023; Cui et al., 2024; Bianchi et al., 2023). One proposed repair is to interleave jailbreak-shaped benign counterexamples, under the hypothesis that these examples teach the model to recognize harmless intent behind adversarial framing.

That hypothesis hides an identification problem. When jailbreak-wrapped benign data reduces over-refusal, at least three explanations remain compatible with the observation. The data could teach presentation-invariant willingness to help with genuinely safe requests. It could teach wrapper-specific adaptation that activates only when a jailbreak-shaped wrapper appears at test time. Alternatively, the benefit could come from adding any extra benign user context to the training distribution. Distinguishing these explanations matters in deployment: a practitioner who believes they purchased generally transferable helpfulness, but actually purchased wrapper-conditional compliance, may mischaracterize both the capabilities and the safety boundary of the adapted model.

We study this problem with a same-core, same-

target controlled training trial. For each request, we construct a dewrapped literal core (D), the identical core inside a jailbreak-shaped wrapper (W), and one audited helpful target valid under both presentations. Following the contrast-set tradition (Gardner et al., 2020; Röttger et al., 2023), only presentation varies; request content, supervised target tokens, hard harmful examples, optimizer slots, initialization, and update count are held fixed across the benign arms. An earlier analysis compared W with D and interpreted the absence of an incremental direct-presentation benefit as evidence of no transfer. This comparison was insufficient: D also receives benign supervision, so a null $W - D$ contrast says nothing about whether benign supervision itself transfers. It isolates only the incremental value of adding a jailbreak wrapper around that supervision.

We close this gap with a normalization-matched harmful-only control H_{norm} . This arm sees the same hard harmful examples, number of optimizer slots and updates, initialization, and total supervised-token denominator as the benign arms, but receives no benign supervision. We also compare it with a slot placebo, $H_{\text{slotplacebo}}$, to test whether the altered loss denominator itself explains the result. These controls separately identify two questions that the original comparison conflated: $D - H_{\text{norm}}$ and $W - H_{\text{norm}}$ measure whether benign supervision transfers at all, while $W - D$ measures whether wrapping that supervision provides an additional benefit.

The result refutes the original no-transfer interpretation. Relative to H_{norm} , both D and W produce large positive improvements in direct-core fulfillment separation: +2.40 and +2.22 log-odds, respectively, with both intervals far from zero. The normalization-matched and slot-placebo harmful-only arms are statistically indistinguishable in every reported matched comparison, so the effect is not explained by the denominator control. What survives is a narrower result: the incremental advantage of W over D depends on test presentation. It is nonpositive on direct cores and positive on wrapped cores, while coexisting with substantial transfer from both benign arms. This behavioral interaction does not establish that the model learned surface form instead of intent; those explanations are not mutually exclusive and are not mechanistically disentangled here.

Evaluation is itself a real-world pitfall in this setting. Cross-judge fulfillment agreement is ex-

cellent on direct prompts but falls sharply on wrapped prompts, exactly where the incremental wrapper effect is measured. We therefore rescore a frozen matched audit with a qualified cross-family Llama-3.1-70B-AWQ-INT4 judge (Dubey et al., 2024; Lin et al., 2023), in addition to the primary Qwen2.5-7B-Instruct judge (Yang et al., 2024). Transfer, the normalization control, and the presentation-dependent $W - D$ pattern reproduce under both judges. A wrapper-stripped evaluator-context ablation further shows that the interaction attenuates but does not disappear when the judge cannot see the wrapper. We consequently describe the result as a two-automatic-judge-qualified behavioral pattern, not a latent-mechanism result. We release the same-core triples, adapters and audits, wrapper-shown and wrapper-stripped judge outputs, cross-judge concordance tables, and analysis code.

2 Related Work

Hard-example tuning and overrefusal. Our study extends the hard-example safety-tuning recipe of Gupta et al. (2026), which mines prompts by how often a target model’s rollouts are judged harmful, trains on its non-jailbroken rollouts, and reduces jailbreak success at the cost of severe overrefusal. That work repairs overrefusal by interleaving adversarially framed benign counterexamples, hypothesizing that these teach harmless intent. It does not hold benign source, core request, target, order, and length fixed, nor does it contrast adversarial framing with a length-matched neutral frame or a harmful-only reference. We provide those controls. More broadly, exaggerated refusal is a known cost of safety supervision (Bianchi et al., 2023; Röttger et al., 2023; Cui et al., 2024). Complementary work attributes overrefusal to keyword or surface shortcuts rather than semantic distinctions (Shi et al., 2024; Wu et al., 2025). This motivates our framing, but we test behavioral transfer rather than the underlying mechanism.

Jailbreak wrappers and presentation dependence. The wrappers we emulate arise from in-the-wild jailbreak ecosystems (Shen et al., 2023), persuasion-based reframings (Zeng et al., 2024), optimization-based attacks (Zou et al., 2023), and WildTeaming (Jiang et al., 2024). The latter’s contrastive vanilla/adversarial and harmful/benign design directly motivates our same-core triples and

target-conditional fragility analysis. Presentation sensitivity is also structurally plausible: Wei et al. (2023) attribute jailbreaks to competing objectives and mismatched generalization, Qi et al. (2024) show that safety alignment is often concentrated in early output tokens, Arditi et al. (2024) localize refusal behavior to a residual-stream direction, and Addepalli et al. (2024) find limited generalization even across semantically related prompts. Our results are compatible with these accounts but do not distinguish among them.

Benign supervision and evaluation. Benign fine-tuning can itself degrade safety, unintentionally (Qi et al., 2023) or as a deliberate few-shot attack (Xie et al., 2025), and task-arithmetic methods have been proposed for subsequent realignment (Bhardwaj et al., 2024). This literature makes a harmful-only anchor necessary: without H_{norm} , continued benign fine-tuning is confounded with the presentation assigned to that benign data. For evaluation, we adopt binary substantive fulfillment and qualification-gated LLM judging in the style of SORRY-Bench and Harm-Bench (Xie et al., 2024; Mazeika et al., 2024), while accounting for known judge biases (Zheng et al., 2023; Panickssery et al., 2024). Estimation uses standard difference-in-differences logic (Angrist and Pischke, 2008), Jeffreys-smoothed log-odds (Brown et al., 2001), crossed bootstrap intervals (Efron and Tibshirani, 1986), Holm correction (Holm, 1979), and κ reliability (McHugh, 2012); we claim no statistical novelty.

3 Method

Same-core triples. For each request, we build a triple (D, W, T) . D is the literal harmless core, W places the identical core inside a jailbreak-shaped wrapper, and T is one audited helpful target valid under both presentations. Training arms differ only in which presentation is paired with T . A tokenizer-length-matched neutral arm N' places the same core inside a non-jailbreak frame. Prompt-token equality is verified for 260/260 benign items and 520/520 interleaved entries under the Qwen tokenizer. All training tensors are hashed and audited per seed.

Harmful-only controls. H_{norm} receives everything the benign arms receive except benign supervision. Per seed, it uses 400 optimizer slots and 50 updates, with 200 hard harmful examples

supervised once and 200 fully masked placeholders. Harmful-token cross-entropy is divided by the harmful plus virtual-benign target-token count: 5,925 harmful target tokens and 11,674 virtual normalization tokens produce a 17,599-token denominator identical to D and W . The virtual tokens carry no benign text, labels, token IDs, or gradients. Thus, $D - H_{\text{norm}}$ and $W - H_{\text{norm}}$ measure the value of benign supervision while holding the harmful-only training regime fixed.

The control ladder includes repeated harmful examples, a slot-matched masked placebo $H_{\text{slotplacebo}}$, and H_{norm} . Repeating harmful examples changes harmful-example exposure and is therefore contextual rather than the clean transfer anchor. The slot-placebo comparison tests whether matching slots and the loss denominator, without benign content, can itself move the endpoint. The normalization-matched arm is the primary reference because it changes neither harmful-example exposure nor the supervised-token denominator relative to the benign arms.

Identification logic. For arm a , source g , and presentation p , let $\Psi_{a,g,p}$ denote fulfillment separation. The two transfer contrasts are

$$T_{D,g,p} = \Psi_{D,g,p} - \Psi_{H_{\text{norm}},g,p}, \quad T_{W,g,p} = \Psi_{W,g,p} - \Psi_{H_{\text{norm}},g,p}.$$

The incremental wrapper contrast is $I_{g,p} = \Psi_{W,g,p} - \Psi_{D,g,p} = T_{W,g,p} - T_{D,g,p}$. Presentation moderation is $M_g = I_{g,\text{wrapped}} - I_{g,\text{direct}}$. Consequently, $I_{g,\text{direct}} \approx 0$ cannot imply $T_D \approx T_W \approx 0$: large common transfer cancels from the treated-versus-treated comparison. This algebra is the central correction to the earlier interpretation.

For estimation, let q be the Jeffreys-smoothed fulfillment probability (Brown et al., 2001) and $\delta_a = \text{logit } q_a - \text{logit } q_{\text{base}}$. We compute category-macro separation as

$$\Psi_{a,g,p} = \text{mean}_{\text{category}} (\delta_{a,\text{safe},p} - \delta_{a,\text{harm},p}).$$

Intervals use 10,000–20,000 crossed seed-and-pair cluster-bootstrap replicates (Efron and Tibshirani, 1986); source tests receive Holm correction (Holm, 1979).

4 Experimental Setup

Models and recipes. Subject models are Qwen2.5-3B-Instruct (Yang et al., 2024) and Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), adapted with LoRA (Hu et al., 2021). We

run a 24-seed primary Qwen panel, an 8-seed alternative-Qwen recipe, and a 24-seed Mistral panel; all pass the frozen headroom and floor gates. The study is LoRA-only, so generality to full fine-tuning, preference optimization, or RLHF remains untested (Rafailov et al., 2023). Appendix A gives the complete experimental record available from the frozen run summaries.

Judges and endpoint. We use two independently qualified automatic judges under an identical fulfillment/refusal/harmfulness rubric: primary Qwen2.5-7B-Instruct and cross-family Llama-3.1-70B-AWQ-INT4 (Dubey et al., 2024; Lin et al., 2023). Their qualification-set human-agreement scores are 0.983/0.983 and 0.966/0.966, respectively, both above a frozen 0.90 gate (Xie et al., 2024; Zheng et al., 2023). The cross-family judge guards against self-preference bias (Panickssery et al., 2024). A fulfillment event requires substantive engagement with the core request; a disclaimer followed by a substantive answer counts, while refusal, evasion, or redirection does not. Harmfulness and refusal are recorded separately and are not equated with fulfillment.

Matched cross-judge audit. Training, generation, and Qwen scoring were frozen before the cross-family audit. The audit preserves the seed, replicate, pair, source, and member assignments of the prior all-pair D/W evaluation, which contains 1,200 bundles and 6,400 responses. Within this structure, the newly scored H_{norm} slice contains 800 responses. The Llama judge parsed all 800 without retries; predictions were hashed before analysis, and integrity checks found no changes to protected artifacts. All two-judge claims use this strictly matched slice rather than an unmatched or partially completed panel. Further provenance and the unanalyzed partial-panel attempt are documented in Appendix A.

5 Results

5.1 Benign supervision transfers

Table 1 reports the transfer contrasts. On direct/XSTest presentations, both benign arms improve fulfillment separation over harmful-only tuning by a large positive margin under both judges. $D - H_{\text{norm}}$ is +2.40 in the full 24-seed Qwen panel, +2.62 in the matched-sample Qwen recomputation, and +2.77 in the Llama audit. Correspond-

Contrast	Pres.	Full-24 Q	Match Q	Llama
$D - H_{\text{norm}}$	Direct	+2.40	+2.62	+2.77
	Wrapped	+0.92	+0.75	+0.64
$W - H_{\text{norm}}$	Direct	+2.22	+2.36	+2.34
	Wrapped	+1.55	+1.39	+1.17

Table 1: Transfer contrasts in category-macro Jeffreys-logit separation. “Full-24 Q” is the 24-seed Qwen panel; “Match Q” uses the Qwen judge on the sample matched to the Llama audit. All twelve recorded 95% bootstrap intervals exclude zero. Exact available direct-presentation intervals appear in Table 7.

Arm	Judge	Direct		Wrapped	
		Safe	Harm	Safe	Harm
D	Qwen	0.810	0.070	—	—
W	Qwen	0.750	0.050	—	—
H_{norm}	Qwen	0.215	0.000	0.080	0.020
H_{norm}	Llama	0.205	0.000	0.060	0.020

Table 2: Raw safe/harmful fulfillment rates. The harmful-only arm is near floor under both judges. The D and W entries are the available direct-presentation primary-judge rates; unavailable cells are not reconstructed.

ing $W - H_{\text{norm}}$ estimates are +2.22, +2.36, and +2.34. Transfer is smaller but remains positive on wrapped presentations: $D - H_{\text{norm}}$ is +0.92/ +0.75/ +0.64, and $W - H_{\text{norm}}$ is +1.55/ +1.39/ +1.17. All twelve recorded transfer intervals exclude zero.

The arm-level rates make the behavioral route concrete (Table 2). Raw direct safe/harmful fulfillment is 0.81/0.07 for D , 0.75/0.05 for W , and 0.215/0.000 for H_{norm} under the primary judge. Harmful-only tuning suppresses fulfillment of both safe and harmful requests. Benign supervision restores much of the safe-core fulfillment while harmful-core fulfillment remains near floor. These results directly refute the no-transfer interpretation: the transfer common to D and W was invisible in their difference.

5.2 The control ladder rules out a denominator artifact

Figure 1 compares both benign arms against three harmful-only controls. The rows are control types, not request categories: repeated harmful examples, a slot-matched masked placebo, and normalization-matched harmful-only training, with 24 seeds per row. Every point is a category-macro safe-minus-harmful separation

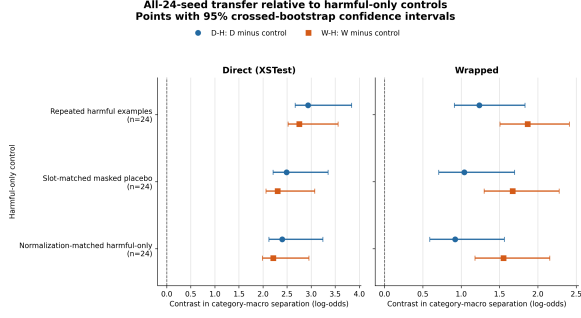


Figure 1: All-24-seed transfer relative to three harmful-only controls. Rows identify the control arm (24 seeds each); columns separate direct/XSTest and wrapped presentations. Blue and orange points are D minus control and W minus control in category-macro safe-minus-harmful separation (log-odds); bars are 95% crossed-bootstrap intervals. Positive values favor benign supervision. The two panels use different horizontal limits.

Setting	Full-crossed Qwen	Llama audit
Direct	+0.091 [-0.032, 0.246]	+0.040 [-0.180, 0.387]
Wrapped	+0.120 [-0.004, 0.260]	—
Moderation	+0.029 [-0.168, 0.214]	—

Table 3: $H_{\text{norm}} - H_{\text{slotplacebo}}$ contrasts. Every reported 95% interval includes zero. The Llama audit contains a direct comparison only; missing comparisons are not inferred.

contrast. Transfer is positive against each control and is larger on direct than wrapped presentations. The panels use different horizontal ranges, so their relative magnitude should be read from the axis values rather than visual bar length. Against the primary H_{norm} anchor, the figure recovers the direct +2.40/ +2.22 and wrapped +0.92/ +1.55 estimates reported above.

The more targeted placebo comparison is null (Table 3). Full-crossed Qwen differences between H_{norm} and $H_{\text{slotplacebo}}$ are +0.091 [-0.032, 0.246] on direct prompts, +0.120 [-0.004, 0.260] on wrapped prompts, and +0.029 [-0.168, 0.214] for moderation. The cross-family direct-audit difference is similarly small, +0.040 [-0.180, 0.387]. Every reported interval includes zero. The matched denominator and masked slots therefore do not explain the transfer attributed to benign content.

5.3 The wrapper’s incremental advantage is presentation-dependent

With transfer established, $W - D$ answers the narrower question for which it is identified: does

Setting	Matched Qwen	Llama
Direct	-0.257 [-0.572, 0.119]	-0.432 [-0.826, 0.069]
Wrapped	+0.639 [0.167, 1.136]	+0.535 [-0.009, 1.125]
Moderation	+0.895 [0.273, 1.476]	+0.967 [0.197, 1.648]

Table 4: Incremental wrapper contrast $W - D$. Moderation is wrapped $W - D$ minus direct $W - D$. Values are category-macro Jeffreys-logit effects with 95% matched bootstrap intervals.

wrapping benign supervision add anything beyond unwrapped benign supervision, and does that increment depend on test presentation? Table 4 and Figure 2 show the same qualitative pattern under both judges. On direct presentations, $W - D$ is nonpositive: -0.26 under Qwen and -0.43 under Llama, with both intervals including zero. On wrapped presentations, the point estimates are positive: +0.64 [0.17, 1.14] under Qwen and +0.54 [-0.01, 1.13] under Llama. The wrapped-minus-direct moderation is positive under both judges, +0.90 [0.27, 1.48] and +0.97 [0.20, 1.65].

Thus, wrapping benign training examples provides no detectable incremental advantage when evaluation uses literal cores, but it provides an incremental advantage when a wrapper is present at evaluation. The statement concerns the interaction of training and test presentation. It does not imply that W fails to transfer; Section 5.1 shows the opposite. Nor does it identify whether the transferred component represents intent learning, semantic generalization, or another shared consequence of benign supervision.

5.4 Judge reliability and evaluator context

The response-level reliability screen in Table 5 explains why cross-family replication is necessary. On direct presentations, fulfillment agreement is 0.971 with $\kappa = 0.939$. On wrapped presentations, it falls to 0.788 with $\kappa = 0.453$. Agreement is high on the matched H_{norm} audit overall (0.990, $\kappa = 0.928$) and within its wrapped subset (0.985, $\kappa = 0.826$), largely because both judges place this arm near fulfillment floor. Aggregate replication and response-level agreement are distinct: close effect estimates do not imply item-wise concordance, and the wrapped-prompt disagreement remains a material limitation.

Figure 3 provides the matched aggregate-effect audit. Both judges recover strongly positive $D - H_{\text{norm}}$ and $W - H_{\text{norm}}$ transfer on direct/XSTest presentations, smaller positive transfer on wrapped

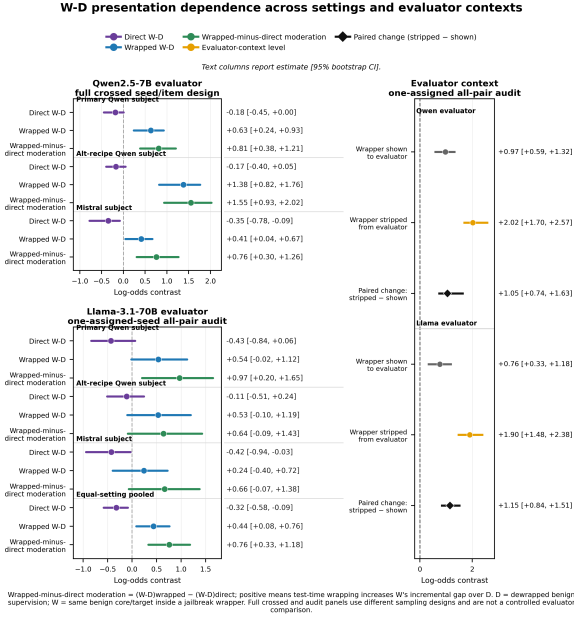


Figure 2: $W - D$ presentation dependence across settings and evaluator contexts. Purple, blue, and green points denote direct $W - D$, wrapped $W - D$, and moderation (wrapped minus direct); horizontal bars are 95% uncertainty intervals. The sign summary concerns point estimates, not universal interval exclusion of zero. In the right panel, identical responses are judged with the wrapper shown or stripped; the black contrast is stripped minus shown. For both judges, moderation falls from approximately +2 log-odds with the wrapper shown to approximately +1 when stripped, but remains positive.

presentations, near-zero $H_{\text{norm}} - H_{\text{slotplacebo}}$, and a nonpositive direct but positive wrapped $W - D$ contrast. The intervals overlap closely at this aggregate level. This is a one-assigned-seed all-pair automatic-judge replication, not a formal judge-equivalence test; the response-level statistics in Table 5 remain the appropriate evidence about item-wise reliability. The length-matched neutral arm N' does not survive this cross-judge screen and is therefore retained only as a judge-dependent exploratory signal.

To test whether wrapped-prompt results merely reflect a judge rewarding visible jailbreak language, we rescore identical model responses after stripping the wrapper from the evaluator’s context (right panel of Figure 2). Moderation attenuates from approximately +2 log-odds when the evaluator sees the wrapper to approximately +1 when it does not; the paired stripped-minus-shown changes are correspondingly negative. Nevertheless, moderation remains positive and ordered con-

Presentation / panel	Raw agreement	κ
Direct (all-pair)	0.971	0.939
Wrapped (all-pair)	0.788	0.453
H_{norm} overall	0.990	0.928
H_{norm} wrapped	0.985	0.826

Table 5: Response-level fulfillment agreement between the Qwen and Llama judges. Aggregate effect replication should not be confused with the substantially weaker item-level agreement on wrapped prompts.

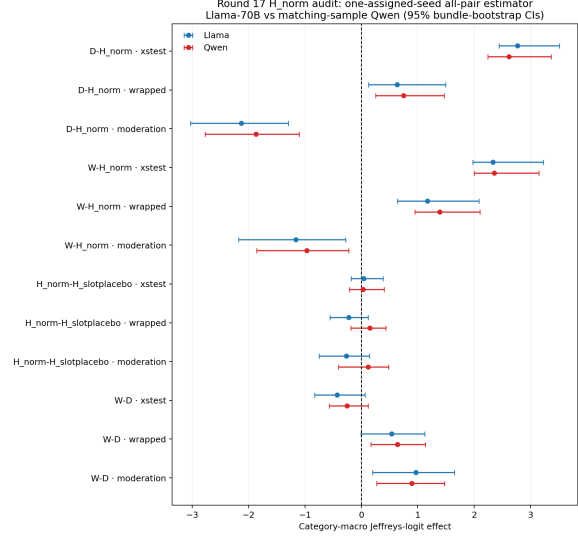


Figure 3: Matched H_{norm} aggregate-effect audit: Llama-3.1-70B-AWQ-INT4 (blue) versus matching-sample Qwen2.5-7B-Instruct (red). XSTest denotes the direct presentation. Rows cover transfer, placebo, and $W - D$ contrasts under direct, wrapped, and moderation settings; points are category-macro Jeffreys-logit effects and bars are 95% bundle-bootstrap intervals. Positive transfer reproduces, the placebo remains near zero, and $W - D$ changes from nonpositive direct to positive wrapped. This is aggregate replication, not response-level judge equivalence.

sistently for both judges. We therefore interpret the interaction as behaviorally attributable but evaluator-sensitive: it is not wholly created by wrapper recognition in the judge, yet the wrapper shown to the judge inflates its apparent magnitude.

5.5 Boundary analyses: safety and target fragility

Safety. We make no safety-improvement claim, and the data do not support one. Fulfillment measures substantive engagement, not whether the resulting content is harmful. No evaluated setting passes the frozen joint 3-point noninferiority gate under either judge. Moreover, the direction of wrapped harmful-request fulfillment reverses

across judges and is therefore unsuitable as evidence of a safety effect. Conventional diagnostics from SORRY-Bench (Xie et al., 2024), OR-Bench-Hard (Cui et al., 2024), and XSTest-safe (Röttger et al., 2023) are included in the released results for completeness, not as support for improved safety.

Target-conditional fragility. The WildJailbreak analysis (Jiang et al., 2024) tests sensitivity to the supervised target rather than transfer. Replacing the original target with a generated canonical target changes the matched moderation from -0.96 to $+2.82$. This sign reversal shows that wrapped-presentation estimates can depend strongly on target construction. Together with the evaluator-context ablation and the wrapped-prompt agreement collapse, it motivates our narrow claim hierarchy: direct-presentation transfer is the most stable result; presentation moderation reproduces but is evaluator-sensitive; target-specific wrapped estimates should not be treated as general safety effects.

6 Conclusion

We tested a strong interpretation, that jailbreak-wrapped benign supervision teaches only wrapper-specific adaptation without transfer, and the missing harmful-only anchor refutes it. Relative to normalization-matched harmful-only tuning, dewrapped and wrapped benign supervision both produce large positive improvements in direct-core fulfillment separation. The normalization-matched and slot-placebo controls are indistinguishable in the reported comparisons, so the effect is not explained by loss normalization. Benign supervision transfers.

The surviving result is narrower: the incremental advantage of placing a jailbreak wrapper around benign supervision is presentation-dependent. It is nonpositive on direct cores and positive on wrapped cores, even though both benign arms transfer substantially. This pattern reproduces under two qualified cross-family automatic judges and survives, with attenuation, when the evaluator cannot see the wrapper. It does not identify a latent intent mechanism and does not demonstrate improved safety.

The broader methodological lesson is portable to safety fine-tuning and other controlled adaptation studies. A treated-versus-treated comparison cannot establish that the supervision shared by both arms has no effect. A supervision-free,

design-matched anchor is required before interpreting a null incremental contrast as absence of transfer. Our scope remains limited to the evaluated FlowGPT-family presentations, two subject-model families, two recipe variants, and LoRA adaptation. Generalization across wrapper families, model scales, full fine-tuning, and preference-based alignment remains open (Rafailov et al., 2023). Future work should complete a full cross-family panel, execute the prepared blinded human audit, and test whether the behavioral interaction corresponds to a mechanistic refusal direction (Arditi et al., 2024; Qi et al., 2024).

References

- Sravanti Addepalli, Yerram Varun, Arun Suggala, Karthikeyan Shanmugam, and Prateek Jain. 2024. Does safety training of llms generalize to semantically related natural prompts? *Preprint*, arXiv:2412.03235.
- Joshua D Angrist and Jörn-Steffen Pischke. 2008. Mostly harmless econometrics: An empiricist’s companion.
- Andy Ardit, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Rimskey, Wes Gurnee, and Neel Nanda. 2024. Refusal in language models is mediated by a single direction. *ArXiv*, abs/2406.11717.
- Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, et al. 2022. Constitutional ai: Harmlessness from ai feedback. *ArXiv*, abs/2212.08073.
- Rishabh Bhardwaj, Do Duc Anh, and Soujanya Poria. 2024. Language models are homer simpson! safety re-alignment of fine-tuned language models through task arithmetic. *Preprint*, arXiv:2402.11746.
- Federico Bianchi, Mirac Suzgun, Giuseppe Attanasio, Paul Röttger, Dan Jurafsky, Tatsunori Hashimoto, and James Zou. 2023. Safety-tuned llamas: Lessons from improving the safety of large language models that follow instructions. *ArXiv*, abs/2309.07875.
- L. Brown, T. T. Cai, and A. Dasgupta. 2001. Interval estimation for a binomial proportion. *Statistical Science*, 16:101–133.
- Justin Cui, Wei-Lin Chiang, Ion Stoica, and ChoJui Hsieh. 2024. Or-bench: An over-refusal benchmark for large language models. *arXiv (Cornell University)*.
- Abhimanyu Dubey, Abhinav Jauhri, et al. 2024. The llama 3 herd of models.
- B. Efron and R. Tibshirani. 1986. Bootstrap methods for standard errors, confidence intervals, and other measures of statistical accuracy. *Statistical Science*, 1:54–75.

560	Matt Gardner, Yoav Artzi, Jonathan Berant, Ben Bo-	Xiangyu Qi, Ashwinee Panda, Kaifeng Lyu, Xiao Ma,	616
561	gin, Sihao Chen, Dheeru Dua, Yanai Elazar, Ananth	Subhrajit Roy, Ahmad Beirami, Prateek Mittal, and	617
562	Gottumukkala, Nitish Gupta, Hannaneh Hajishirzi,	Peter Henderson. 2024. Safety alignment should	618
563	Gabriel Ilharco, Daniel Khashabi, Kevin Lin, Jiang-	be made more than just a few tokens deep. <i>ArXiv</i> ,	619
564	ming Liu, Nelson F. Liu, Phoebe Mulcaire, Qiang	abs/2406.05946.	620
565	Ning, Sameer Singh, Noah A. Smith, and 4 others.		
566	2020. Evaluating models' local decision boundaries	Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen,	621
567	via contrast sets. pages 1307–1323.	Ruoxi Jia, Prateek Mittal, and Peter Henderson.	622
		2023. Fine-tuning aligned language models com-	623
568	Prakhar Gupta, Garv Shah, and Donghuan Zhang. 2026.	promises safety, even when users do not intend to!	624
569	Self-mined hardness for safety fine-tuning. <i>ArXiv</i> ,	<i>ArXiv</i> , abs/2310.03693.	625
570	abs/2605.03226.		
571	S. Holm. 1979. A simple sequentially rejective multi-	Rafael Rafailov, Archit Sharma, E. Mitchell, S. Er-	626
572	ple test procedure. <i>Scandinavian Journal of Statis-</i>	mon, Christopher D. Manning, and Chelsea Finn.	627
573	<i>tics</i> , 6:65–70.	2023. Direct preference optimization: Your lan-	628
		guage model is secretly a reward model. <i>ArXiv</i> ,	629
574	J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu,	abs/2305.18290.	630
575	Yuanzhi Li, Shean Wang, and Weizhu Chen. 2021.		
576	Lora: Low-rank adaptation of large language mod-	Paul Röttger, Hannah Rose Kirk, Bertie Vidgen,	631
577	els. <i>ArXiv</i> , abs/2106.09685.	Giuseppe Attanasio, Federico Bianchi, and Dirk	632
		Hovy. 2023. Xstest: A test suite for identifying exag-	633
578	Albert Qiaochu Jiang, Alexandre Sablayrolles, Arthur	gerated safety behaviours in large language models.	634
579	Mensch, Chris Bamford, Devendra Singh Chap-	<i>ArXiv</i> , abs/2308.01263.	635
580	lot, Diego de Las Casas, Florian Bressand, Gi-		
581	anna Lengyel, Guillaume Lample, Lucile Saulnier,	Xinyue Shen, Zeyuan Chen, M. Backes, Yun Shen, and	636
582	Lélio Renard Lavaud, M. Lachaux, Pierre Stock,	Yang Zhang. 2023. "do anything now": Characteriz-	637
583	Teven Le Scao, Thibaut Lavril, Thomas Wang, Tim-	ing and evaluating in-the-wild jailbreak prompts on	638
584	othée Lacroix, and William El Sayed. 2023. Mistral	large language models. <i>Proceedings of the 2024 on</i>	639
585	7b. <i>ArXiv</i> , abs/2310.06825.	<i>ACM SIGSAC Conference on Computer and Com-</i>	640
		<i>munications Security</i> .	641
586	Liwei Jiang, Kavel Rao, Seungju Han, Allyson Et-	Chenyu Shi, Xiao Wang, Qiming Ge, Songyang Gao,	642
587	tinger, Faeze Brahman, Sachin Kumar, Niloofar	Xianjun Yang, Tao Gui, Qi Zhang, Xuanjing Huang,	643
588	Miresghallah, Ximing Lu, Maarten Sap, Yejin	Xun Zhao, and Dahua Lin. 2024. Navigating	644
589	Choi, and Nouha Dziri. 2024. Wildteaming at scale:	the overkill in large language models. <i>ArXiv</i> ,	645
590	From in-the-wild jailbreaks to (adversarially) safer	abs/2401.17633.	646
591	language models. <i>ArXiv</i> , abs/2406.18510.		
592	Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang,	Alexander Wei, Nika Haghtalab, and J. Steinhardt.	647
593	Xingyu Dang, and Song Han. 2023. Awq:	2023. Jailbroken: How does llm safety training fail?	648
594	Activation-aware weight quantization for on-device	<i>ArXiv</i> , abs/2307.02483.	649
595	llm compression and acceleration. <i>GetMobile: Mo-</i>		
596	<i>bile Computing and Communications</i> , 28:12 – 17.	Xiaorui Wu, Li Fei, Xiaofeng Mao, Xin Zhang, Zheng	650
597	Mantas Mazeika, Long Phan, Xuwang Yin, Andy	Li, Yuxiang Peng, Chong Teng, Donghong Ji, and	651
598	Zou, Zifan Wang, Norman Mu, Elham Sakhaee,	Zhuang Li. 2025. Evorefuse: Evolutionary prompt	652
599	Nathaniel Li, Steven Basart, Bo Li, David Forsyth,	optimization for evaluation and mitigation of llm	653
600	and Dan Hendrycks. 2024. Harmbench: A standard-	over-refusal to pseudo-malicious instructions.	654
601	ized evaluation framework for automated red team-		
602	ing and robust refusal. pages 35181–35224.	Tinghao Xie, Xiangyu Qi, Yi Zeng, Yangsibo Huang,	655
603	Marry L. McHugh. 2012. Interrater reliability: the	Udari Madhushani Schwag, Kaixuan Huang, Luxi	656
604	kappa statistic. <i>Biochemia Medica</i> .	He, Boyi Wei, Dacheng Li, Ying Sheng, Ruoxi Jia,	657
605	Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida,	Bo Li, Kai Li, Danqi Chen, Peter Henderson, and	658
606	Carroll L. Wainwright, Pamela Mishkin, Chong	Prateek Mittal. 2024. Sorry-bench: Systematically	659
607	Zhang, S. Agarwal, Katarina Slama, Alex Ray, John	evaluating large language model safety refusal be-	660
608	Schulman, Jacob Hilton, Fraser Kelton, Luke E.	haviors. <i>ArXiv</i> , abs/2406.14598.	661
609	Miller, M. Simens, Amanda Askell, Peter Welin-		
610	der, P. Christiano, Jan Leike, and Ryan J. Lowe.	Zhixin Xie, Xurui Song, and Jun Luo. 2025. Attack via	662
611	2022. Training language models to follow instruc-	overfitting: 10-shot benign fine-tuning to jailbreak	663
612	tions with human feedback. <i>ArXiv</i> , abs/2203.02155.	llms. <i>Preprint</i> , arXiv:2510.02833.	664
613	Arjun Panickssery, Samuel R. Bowman, and Shi Feng.		
614	2024. Llm evaluators recognize and favor their own	Qwen An Yang, Baosong Yang, et al. 2024. Qwen2.5	665
615	generations. <i>ArXiv</i> , abs/2404.13076.	technical report. <i>ArXiv</i> , abs/2412.15115.	666
		Yi Zeng, H. Lin, Jingwen Zhang, Diyi Yang, Ruoxi	667
		Jia, and Weiyan Shi. 2024. How johnny can per-	668
		sue llms to jailbreak them: Rethinking persuasion	669
		to challenge ai safety by humanizing llms. <i>ArXiv</i> ,	670
		abs/2401.06373.	671

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhonghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, E. Xing, Haoteng Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. *ArXiv*, abs/2306.05685.

Andy Zou, Zifan Wang, J. Kolter, and Matt Fredrikson. 2023. Universal and transferable adversarial attacks on aligned language models. *ArXiv*, abs/2307.15043.

A Full Experimental and Reproducibility Record

Component	Frozen setting
Primary subject	Qwen2.5-3B-Instruct
Cross-family subject	Mistral-7B-Instruct-v0.3
Adaptation	LoRA on attention and MLP projections
Primary panel	24 seeds
Alternative Qwen panel	8 seeds
Mistral panel	24 seeds
Optimizer slots per seed	400
Updates per seed	50
Hard harmful examples	200, supervised once
Masked placeholders	200
Harmful target tokens	5,925
Virtual normalization tokens	11,674
Matched denominator	17,599 target tokens
Primary judge	Qwen2.5-7B-Instruct
Cross-family judge	Llama-3.1-70B-AWQ-INT4
Bootstrap replicates	10,000–20,000

Table 6: Experimental values exposed by the frozen run summaries and used in this manuscript. Detailed optimizer, LoRA-rank, and decoding configurations are preserved in the released machine-readable run configurations rather than reconstructed from incomplete prose records.

Subject models and adaptation. The subject models are Qwen2.5-3B-Instruct (Yang et al., 2024) and Mistral-7B-Instruct-v0.3 (Jiang et al., 2023). LoRA modules (Hu et al., 2021) are applied to attention and MLP projection matrices. The panels contain 24 seeds for the primary Qwen recipe, 8 seeds for the alternative Qwen recipe, and 24 seeds for the Mistral recipe. Each arm (D , W , N' , H_{norm} , and $H_{\text{slotplacebo}}$) shares the hard harmful examples, optimizer-slot structure, initialization policy, and update count, differing only where required by the intervention. Per-seed data audits, adapter hashes, and machine-readable configurations accompany the released artifacts.

Training and normalization. Each seed uses 400 optimizer slots and 50 updates. The harmful-only reference contains 200 hard harmful exam-

ples supervised once and 200 fully masked placeholders. The matched supervised-token denominator is 17,599, consisting of 5,925 harmful target tokens and 11,674 virtual normalization tokens. Virtual tokens contribute only to the denominator and provide no text, token IDs, labels, or gradients. This construction prevents the harmful-only reference from receiving benign content while matching the optimization normalization used by the benign arms.

Neutral-frame token matching. For N' , the neutral frame is verified to exact Qwen prompt-token equality with the wrapped frame on 260/260 benign items and 520/520 interleaved entries. This controls tokenized prompt length but cannot ensure identical optimization dynamics, semantic salience, or judge behavior. Since the neutral-frame result does not pass the cross-judge screen, it is not included in the confirmatory claim hierarchy.

Judges and qualification. The primary judge is Qwen2.5-7B-Instruct (Yang et al., 2024), with qualification-set human agreement 0.983/0.983. The cross-family judge is Llama-3.1-70B-AWQ-INT4 (Dubey et al., 2024; Lin et al., 2023), with agreement 0.966/0.966. Both exceed the frozen 0.90 qualification gate, following qualification-gated LLM evaluation practice (Xie et al., 2024; Zheng et al., 2023; Mazeika et al., 2024). Both use the same fulfillment, refusal, and harmfulness rubric. The cross-family judge addresses, but cannot eliminate, evaluator-family and self-preference concerns (Panickssery et al., 2024).

Outcome definition. A fulfillment event requires substantive engagement with the core request. A response that includes a disclaimer and then provides a substantive answer counts as fulfillment. Refusal, evasion, or redirection does not. Refusal and harmfulness are separate labels. This distinction is central to the bounded safety conclusion: low harmful-request fulfillment can be safety-relevant, but fulfillment alone does not determine whether generated content is harmful.

Estimation. Per-arm probabilities use a Jeffreys Beta($1/2, 1/2$) prior (Brown et al., 2001), avoiding divergent logits in zero-count cells such as the raw H_{norm} harmful-fulfillment cell. Effects are category-macro logit contrasts. Confidence intervals use crossed seed-and-pair cluster bootstrap resampling (Efron and Tibshirani, 1986),

with 10,000–20,000 replicates depending on the frozen analysis. Source-level tests are Holm-corrected (Holm, 1979). Inter-judge reliability uses raw agreement and Cohen’s κ (McHugh, 2012).

Cross-family audit provenance. All model training, response generation, and Qwen scoring predate the cross-family audit. The all-pair audit structure contains 1,200 bundles and 6,400 responses. The newly scored H_{norm} slice contains 800 responses with assignments inherited exactly from that structure. The Llama judge parsed 800/800 items with zero retries. Predictions were hashed before analysis, and integrity checks confirmed that no protected prior artifact changed.

A full 38,400-response Llama H_{norm} panel was also attempted. Of the 37,600 items not already present in the matched audit, 12,000 were scored before a two-hour execution limit stopped the run. The completed prefix is not a representative sample, so no partial-panel statistics are reported. A bookkeeping audit corrected an inherited panel-size record of 25,600 to the actual 38,400. This correction affects provenance only, not any reported estimate.

Human evaluation status. A blinded human-audit package was prepared and integrity-checked, but no genuine annotators were available during the run. No human labels were fabricated or simulated. All confirmatory findings are therefore labeled two-automatic-judge-qualified rather than human-validated.

B Complete Aggregate Results

Contrast (direct)	Panel / judge	Estimate [95% CI]
$D - H_{\text{norm}}$	Full-24 Qwen	+2.397 [2.122, 3.246]
	Matched Qwen	+2.615 [2.244, 3.365]
	Llama	+2.768 [2.445, 3.512]
$W - H_{\text{norm}}$	Full-24 Qwen	+2.215 [1.987, 2.955]
	Matched Qwen	+2.359 [2.000, 3.151]
	Llama	+2.336 [1.978, 3.226]
<i>Wrapped-presentation point estimates</i>		
$D - H_{\text{norm}}$	Full / matched / Llama	+0.921 / +0.749 / +0.772
$W - H_{\text{norm}}$	Full / matched / Llama	+1.552 / +1.388 / +1.473

Table 7: Available exact transfer intervals for direct presentations and recorded point estimates for wrapped presentations. The frozen analyses record all corresponding wrapped intervals as excluding zero, but exact wrapped interval endpoints are not reconstructed where they are absent from the run summary.

$W - D$ setting	Matched Qwen	Llama
Direct	−0.257 [−0.572, 0.119]	−0.432 [−0.826, 0.069]
Wrapped	+0.639 [0.167, 1.136]	+0.535 [−0.009, 1.125]
Moderation	+0.895 [0.273, 1.476]	+0.967 [0.197, 1.648]

Table 8: Complete matched incremental wrapper contrasts. Moderation is wrapped minus direct.

Placebo setting	Full-crossed Qwen	Llama audit
Direct	+0.091 [−0.032, 0.246]	+0.040 [−0.180, 0.387]
Wrapped	+0.120 [−0.004, 0.260]	Not available
Moderation	+0.029 [−0.168, 0.214]	Not available

Table 9: Complete reported $H_{\text{norm}} - H_{\text{slotplacebo}}$ comparisons. No unobserved Llama cells are inferred.

Conventional diagnostics. Per-arm results for SORRY-Bench English (Xie et al., 2024), OR-Bench-Hard English (Cui et al., 2024), and XSTest-safe false refusal (Röttger et al., 2023) are included in the released machine-readable result tables. They are secondary diagnostics rather than evidence for a safety improvement. No setting passes the frozen joint 3-point noninferiority gate, and the wrapped harmful-request direction changes across judges.

Transport-model falsification. Global additive and category-conditioned affine transport models fit only to safe members do not reproduce the observed presentation moderation. This rejects a simple scalar or affine shift as a sufficient description of the interaction, but it does not identify a representation-level mechanism. Within-source pair-cluster falsification and unpaired AU-ROC/AUPRC checks are consistent with the clustered-bootstrap conclusions and are included with the analysis artifacts.

C Additional Recipe and Evaluator Analysis

Figure 4 decomposes transfer by evaluator, presentation, recipe, and harmful-only control. Across the primary Qwen configurations, both judges receive large positive direct $D - H$ and $W - H$ contrasts, with smaller wrapped contrasts. Repeated-harmful and slot-placebo controls yield broadly similar primary-recipe patterns. The alternative Qwen and Mistral panels show that positive transfer is not confined to a single subject recipe, while also revealing recipe-specific exceptions to a universal direct-greater-than-wrapped ordering. The figure therefore supports cross-recipe robustness

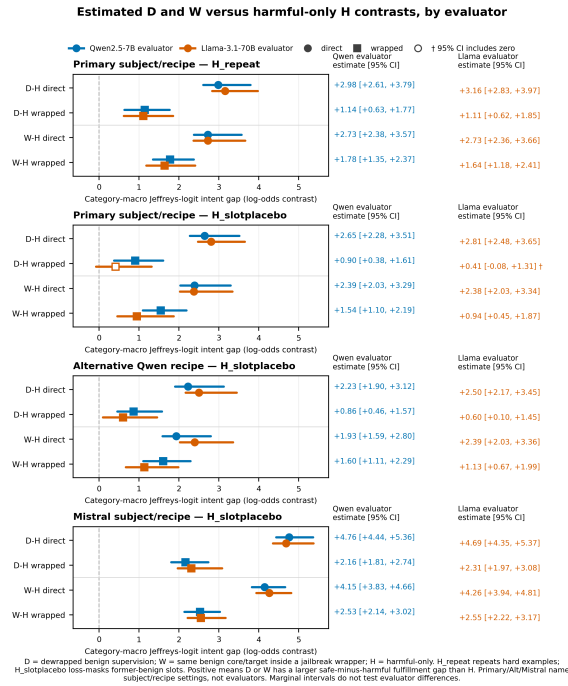


Figure 4: Estimated D and W versus harmful-only controls, separated by evaluator, presentation, subject recipe, and control construction. Panels show the primary recipe with repeated-harmful and slot-placebo controls, the alternative Qwen recipe with the slot placebo, and the Mistral subject/recipe with the slot placebo. Positive category-macro log-odds contrasts favor benign-supervised D or W over the harmful-only arm; horizontal bars are 95% intervals. Most estimates are positive under both evaluators, while some wrapped recipe-specific intervals approach or cross zero. The primary repeated-harmful and slot-placebo patterns are similar, but direct effects are not universally larger in every recipe.

of transfer, not invariance of its exact magnitude.

D Scope, Limitations, and Claim Audit

Corrected interpretation. The manuscript removes the earlier no-transfer and surface-form-instead-of-intent conclusions. The harmful-only anchor shows that both D and W substantially increase direct-core fulfillment separation. Intent learning and surface sensitivity are not mutually exclusive, and the present behavioral design does not identify either as the unique mechanism.

Confirmatory claims. Three findings satisfy the two-automatic-judge standard: benign supervision transfers relative to H_{norm} ; the reported normalization-control comparisons are near zero; and the incremental $W - D$ effect is more positive on wrapped than direct presentations. The neutral-

frame N' result is exploratory because it does not survive the cross-judge screen.

Evaluator limitations. Wrapped-prompt response-level agreement is only 0.788 with $\kappa = 0.453$. Wrapper stripping reduces but does not eliminate the moderation. Thus, the interaction is not wholly an evaluator-context artifact, but its magnitude is evaluator-sensitive. The aggregate forest plot cannot establish judge equivalence because both judges score the same correlated responses and no paired equivalence margin was specified.

Design limitations. Exact token-length matching does not equalize semantic salience or all training dynamics. Neutral-arm degradation may therefore reflect differences that token counts alone cannot remove. The target-conditional sign reversal further shows that wrapper effects can depend strongly on target construction. The study evaluates one held-out FlowGPT-family presentation distribution, two subject-model families, two recipe variants, and LoRA adaptation only. It does not establish generality to other wrapper families, larger models, full-parameter fine-tuning, DPO, or RLHF (Rafailov et al., 2023).

Statistical limitations. The analysis treats seeds and matched request pairs as crossed uncertainty sources and uses category-macro effects to prevent large categories from dominating. With only a small number of recipe settings, recipe heterogeneity should not be interpreted as population-level random-effects evidence. Several wrapped contrasts are close to zero, and adding training seeds alone cannot resolve uncertainty dominated by evaluation items or judge disagreement.

Artifact release. The release contains same-core triples; H_{norm} and $H_{\text{slotplacebo}}$ adapters; per-seed data and token audits; wrapper and source hashes; both judges' wrapper-shown and wrapper-stripped outputs; the reliability screen; cross-judge concordance tables; per-setting clustered effects; the prepared blinded human-audit package; and analysis code. Partial, nonrepresentative Llama-panel outputs are clearly marked and excluded from all reported statistics.